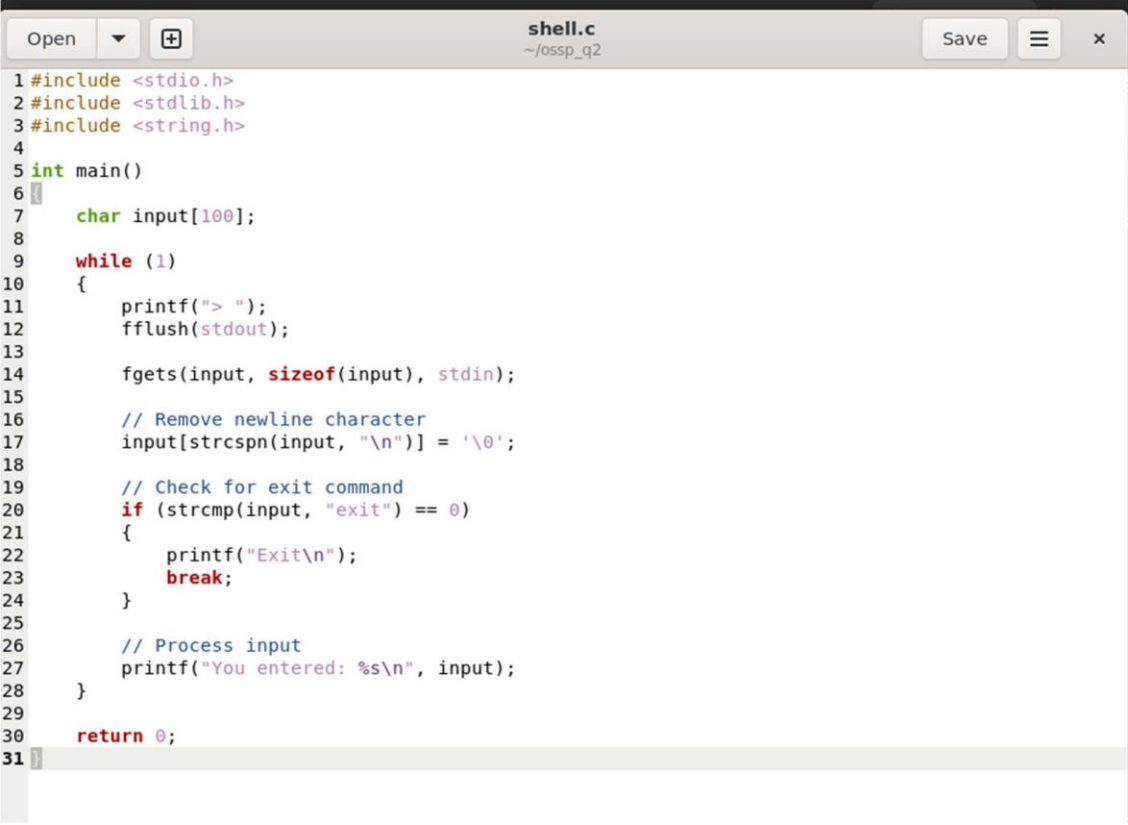


OS Skill – 2

Problem 1



```
1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <string.h>
4
5 int main()
6 {
7     char input[100];
8
9     while (1)
10    {
11        printf("> ");
12        fflush(stdout);
13
14        fgets(input, sizeof(input), stdin);
15
16        // Remove newline character
17        input[strcspn(input, "\n")] = '\0';
18
19        // Check for exit command
20        if (strcmp(input, "exit") == 0)
21        {
22            printf("Exit\n");
23            break;
24        }
25
26        // Process input
27        printf("You entered: %s\n", input);
28    }
29
30    return 0;
31 }
```

```
anirudh@Anirudh:~$ mkdir -p ~/ossp_q2
anirudh@Anirudh:~$ cd ~/ossp_q2
anirudh@Anirudh:~/ossp_q2$ pwd
/home/anirudh/ossp_q2
anirudh@Anirudh:~/ossp_q2$ gedit shell.c
anirudh@Anirudh:~/ossp_q2$ gcc shell.c -o shell
anirudh@Anirudh:~/ossp_q2$ ls
shell  shell.c
anirudh@Anirudh:~/ossp_q2$ ./shell
> hello
You entered: hello
> test command
You entered: test command
> exit
Exit
anirudh@Anirudh:~/ossp_q2$
```

Problem 2

```
1 #include <stdio.h>
2 #include <unistd.h>
3 #include <termios.h>
4
5 #define BUFFER_SIZE 100
6
7 void enable_raw_mode()
8 {
9     struct termios terminal;
10    tcgetattr(STDIN_FILENO, &terminal);
11    terminal.c_lflag &= ~(ICANON | ECHO);
12    tcsetattr(STDIN_FILENO, TCSANOW, &terminal);
13 }
14
15 void disable_raw_mode()
16 {
17     struct termios terminal;
18    tcgetattr(STDIN_FILENO, &terminal);
19    terminal.c_lflag |= (ICANON | ECHO);
20    tcsetattr(STDIN_FILENO, TCSANOW, &terminal);
21 }
22
23 int main()
24 {
25     char buffer[BUFFER_SIZE];
26     int index = 0;
27     char ch;
28
29     enable_raw_mode();
30
31     printf("> ");
32     fflush(stdout);
33 }
```

```
Open keyboard.c ~/ossp_q2 Save
38 fflush(stdout);
39
40 while (1)
41 {
42     read(STDIN_FILENO, &ch, 1);
43
44     /* Handle Enter key */
45     if (ch == '\n' || ch == '\r')
46     {
47         buffer[index] = '\0';
48
49         printf("\nYou entered: %s\n", buffer);
50         printf("Buffer: %s\n", buffer);
51
52         /* Handle exit command */
53         if (index == 4 &&
54             buffer[0] == 'e' &&
55             buffer[1] == 'x' &&
56             buffer[2] == 'i' &&
57             buffer[3] == 't')
58         {
59             printf("Exit\n");
60             break;
61         }
62
63         /* Reset input buffer */
64         index = 0;
65
66         printf("> ");
67         fflush(stdout);
68     }
69
70     /* Handle Backspace */
71     else if (ch == 127 || ch == '\b')
72     {
73         if (index > 0)
74         {
```

```
70         /* Handle Backspace */
71         else if (ch == 127 || ch == '\b')
72         {
73             if (index > 0)
74             {
75                 index--;
76
77                 printf("\b \b");
78                 fflush(stdout);
79             }
80         }
81
82         /* Handle normal keyboard characters */
83         else if (index < BUFFER_SIZE - 1)
84         {
85             buffer[index] = ch;
86             index++;
87
88             printf("%c", ch);
89             fflush(stdout);
90         }
91     }
92
93     disable_raw_mode();
94
95     return 0;
96 }
```

```
anirudh@Anirudh:~/ossp_q2$ gedit shell.c
anirudh@Anirudh:~/ossp_q2$ gedit keyboard.c
anirudh@Anirudh:~/ossp_q2$ gcc keyboard.c -o keyboard
anirudh@Anirudh:~/ossp_q2$ ./keyboard
> hello
You entered: hello
> hello
You entered: hello
> test command
You entered: test command
> exit
You entered: exit
Exit
anirudh@Anirudh:~/ossp_q2$ gedit keyboard.c
anirudh@Anirudh:~/ossp_q2$ gcc keyboard.c -o keyboard
anirudh@Anirudh:~/ossp_q2$ ./keyboard
> hello
You entered: hello
Buffer: hello
> hello
You entered: hello
Buffer: hello
> hello
You entered: hello
Buffer: hello
> |
```